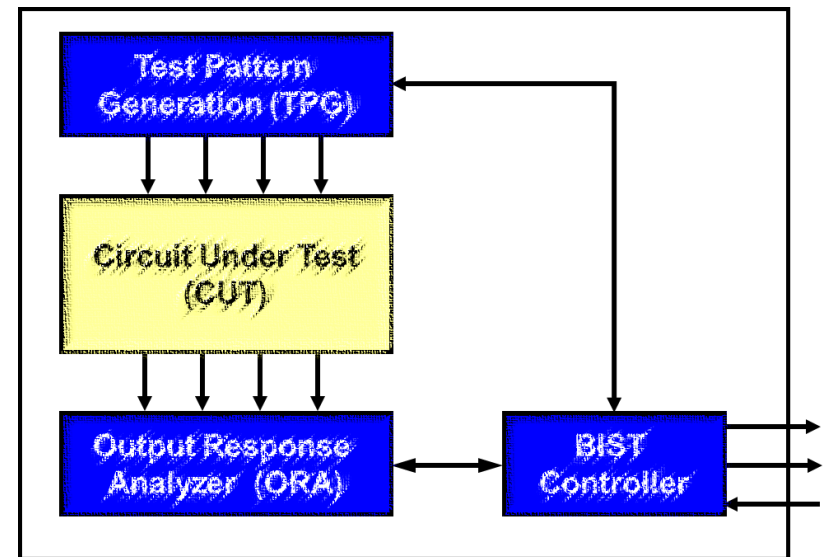


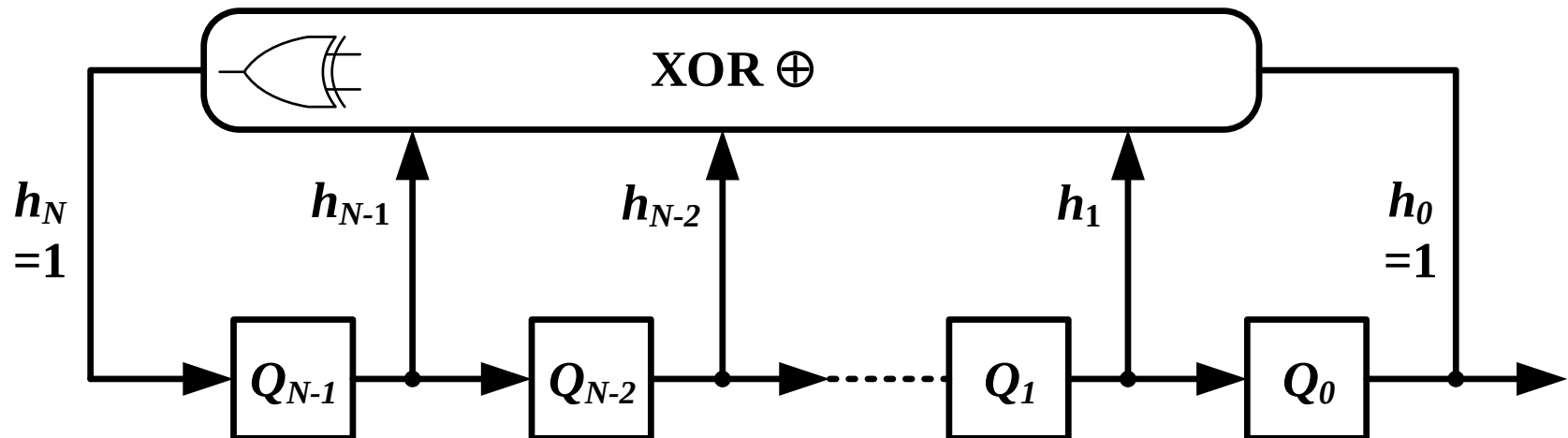
BIST Part1 - TPG

- Introduction
- Test Pattern Generation (TPG)
 - ◆ Deterministic: ROM, Algorithm, Counter
 - ◆ Pseudo Random
 - * Linear Feedback Shift Register, LFSR (1977)
 - Two types of LFSR
 - Design of LFSR
 - * Cellular Automata, CA (1984)



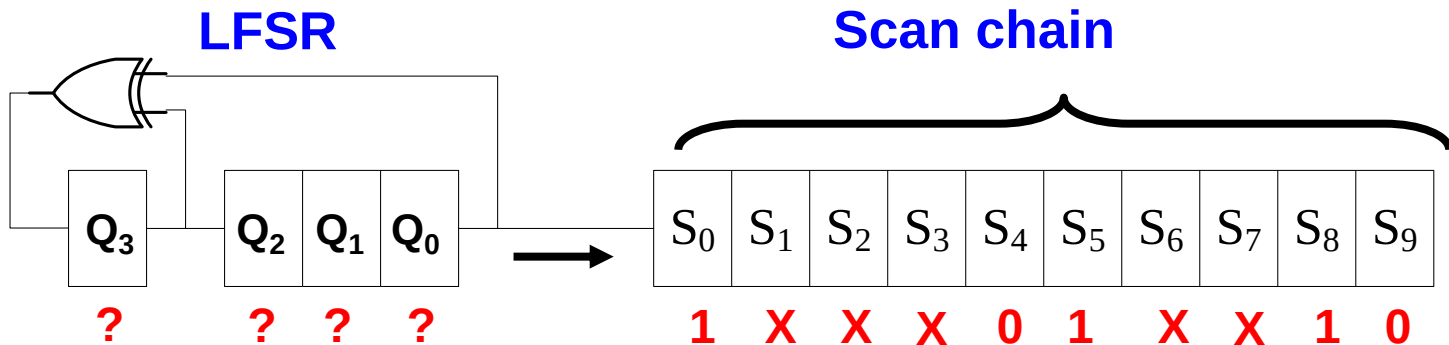
Design of LFSR

- Questions to answer
 - ♦ How to find seed?
 - ♦ What is LFSR degree?
 - ♦ What polynomial?



How to Find a Seed?

- Given LFSR and test pattern, find initial state of LFSR (seed)
- Example:
 - ♦ LFSR feeds scan chain inputs
 - ♦ $S_0 \sim S_9$ are test pattern (specified by ATPG)
 - ♦ $Q_3 \sim Q_0$ are seeds

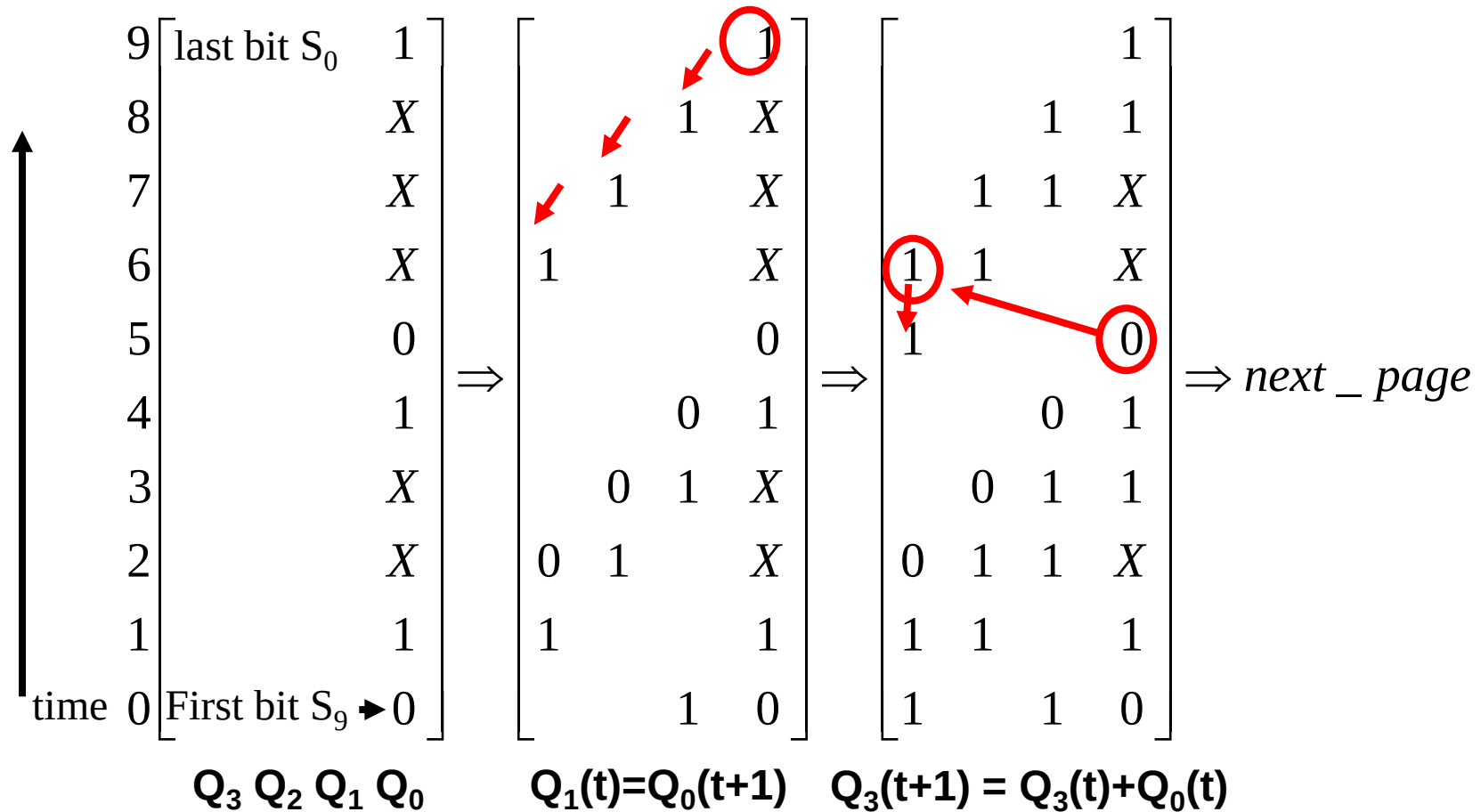
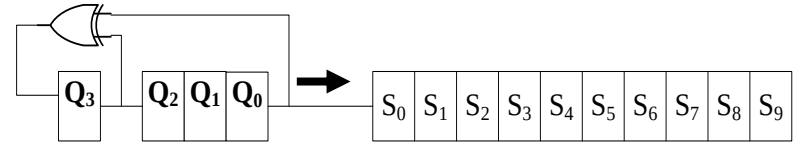


- Two methods:
 - ♦ Cycle-by-cycle tracing
 - ♦ System of linear equations

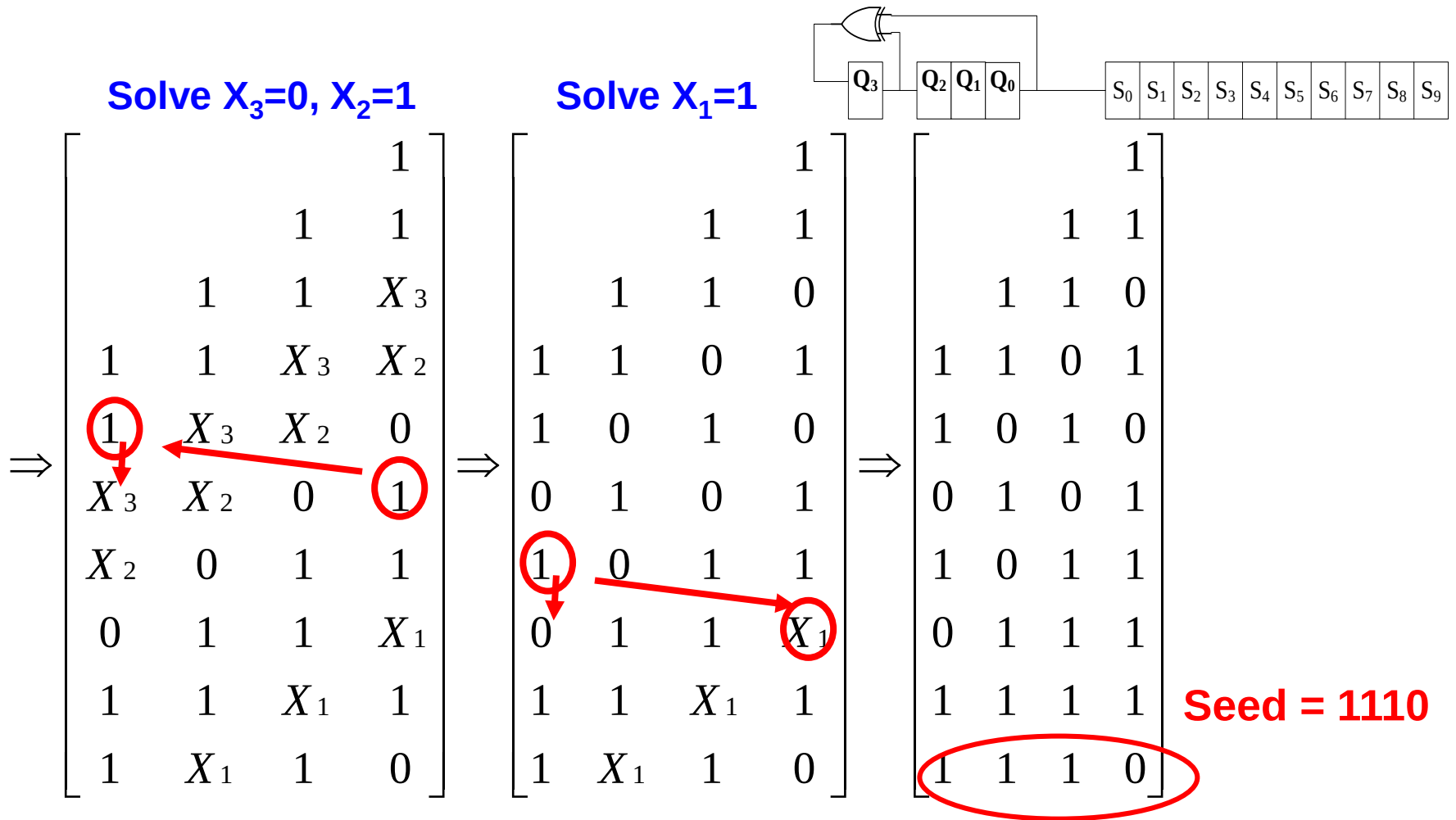
Cycle-by-cycle Tracing (1)

- **Test pattern desired**

♦ $S_0 S_1 \dots S_9 = 1 X X X 0 1 X X 1 0$



Cycle-by-cycle Tracing (2)



This is Slow! Can We Do Better?

System of Linear Equations (1)

- Initial conditions

- ♦ $S_9 = Q_0$

- ♦ $S_8 = Q_1$

- ♦ $S_7 = Q_2$

- ♦ $S_6 = Q_3$

- ♦ $S_n = S_{n+4} \oplus S_{n+1}$

- ♦ $S_5 = S_9 \oplus S_6 = Q_3 \oplus Q_0$

- ♦ $S_4 = S_8 \oplus S_5 = Q_3 \oplus Q_1 \oplus Q_0$

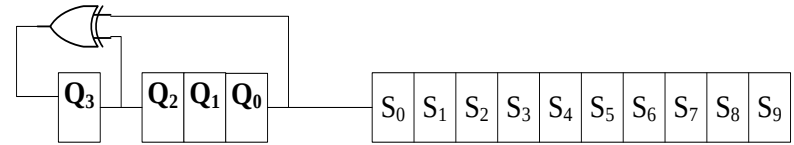
- ♦ $S_3 = S_7 \oplus S_4 = Q_3 \oplus Q_2 \oplus Q_1 \oplus Q_0$

- ♦ $S_2 = S_6 \oplus S_3 = Q_2 \oplus Q_1 \oplus Q_0$

- ♦ $S_1 = S_5 \oplus S_2 = Q_3 \oplus Q_2 \oplus Q_1$

- ♦ $S_0 = S_4 \oplus S_1 = Q_2 \oplus Q_0$

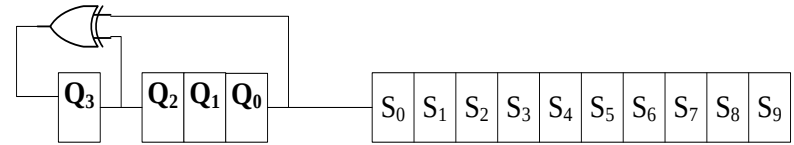
- 4 variables, 10 equations



$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Q_3 \\ Q_2 \\ Q_1 \\ Q_0 \end{bmatrix} = \begin{bmatrix} S_0 = 1 \\ S_1 = X \\ S_2 = X \\ S_3 = X \\ S_4 = 0 \\ S_5 = 1 \\ S_6 = X \\ S_7 = X \\ S_8 = 1 \\ S_9 = 0 \end{bmatrix}$$

System of Linear Equations (2)

- $S_9 = 0 = Q_0$
- $S_8 = 1 = Q_1$
- $S_5 = 1 = Q_0 \oplus Q_3 \Rightarrow Q_3 = 1$
- $S_4 = 0 = Q_3 \oplus Q_1 \oplus Q_0$
 - ♦ consistent !
- $S_0 = 1 = Q_2 \oplus Q_0 \Rightarrow Q_2 = 1$
- Solution :
 - ♦ Seed = $[Q_3 \ Q_2 \ Q_1 \ Q_0] = 1110$

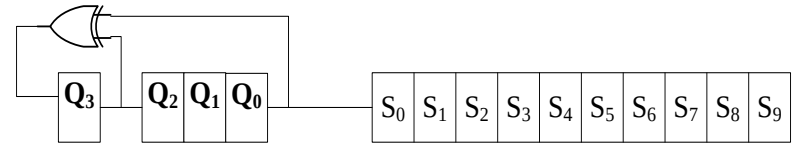


$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Q_3 \\ Q_2 \\ Q_1 \\ Q_0 \end{bmatrix} = \begin{bmatrix} S_0 = 1 \\ S_1 = X \\ S_2 = X \\ S_3 = X \\ S_4 = 0 \\ S_5 = 1 \\ S_6 = X \\ S_7 = X \\ S_8 = 1 \\ S_9 = 0 \end{bmatrix}$$

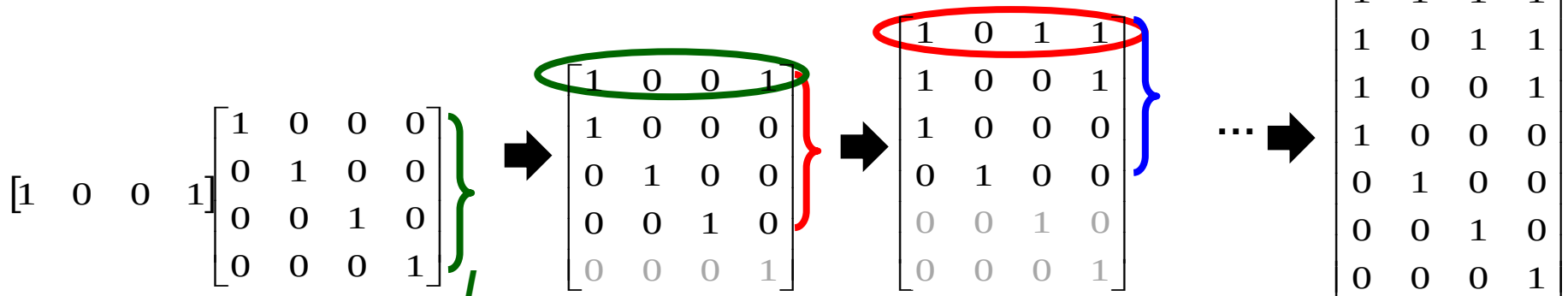
Automate This Process?

Derive Linear Equations (1)

- Start with identity matrix I
 - $t = [h_3 \ h_2 \ h_1 \ h_0]$
- Iteratively bottom up, $k=4 \sim 10$
 - $\text{Row}_k = t \times (\text{Row}_{k-4} \sim \text{Row}_{k-1})$
 - $k++$



- Example
 - $t = [h_3 \ h_2 \ h_1 \ h_0] = [1 \ 0 \ 0 \ 1]$
 - $t \times I_1 = [1 \ 0 \ 0 \ 1]$
 - $t \times I_2 = [1 \ 0 \ 1 \ 1]$
 -
 - Eventually, 10 rows, 4 columns



Solve Linear Equations (1)

- **Gauss-Jordan elimination**, but *Mod-2 addition*
- **Example**
 - ♦ Append specified bits to last column (*Augment matrix*)
 - ♦ Remove useless rows with X
 - ♦ Exchange row 1 with row 2

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & X \\ 0 & 1 & 1 & 1 & X \\ 1 & 1 & 1 & 1 & X \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & X \\ 0 & 1 & 0 & 0 & X \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$


Solve Linear Equations (2)

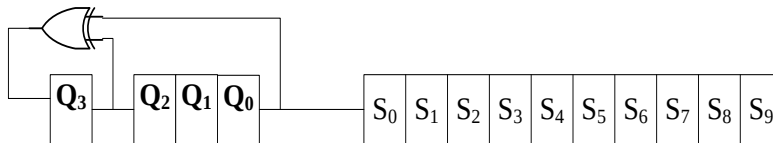
- Pivot column #1 (add row 1 to row 3)
- Pivot column #2 (do nothing)
- Pivot column #3 (add row3 to row1; add row3 to row4)
- Pivot column #4 (add row5 to row1, add row 5 to row2)
- Solution: seed $= [Q_3 \ Q_2 \ Q_1 \ Q_0] = [1 \ 1 \ 1 \ 0]$

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
 \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
 \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
 \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
 \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Summary



- Number of *equations* depends on *number of care bits*
- Number of *variables* = *LFSR degree*
- If more care bits than degree,
 - ♦ May not be solvable every time
 - ♦ This time we are lucky :) 



$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Q_3 \\ Q_2 \\ Q_1 \\ Q_0 \end{bmatrix} = \begin{bmatrix} S_0 = 1 \\ S_1 = X \\ S_2 = X \\ S_3 = X \\ S_4 = 0 \\ S_5 = 1 \\ S_6 = X \\ S_7 = X \\ S_8 = 1 \\ S_9 = 0 \end{bmatrix}$$

Can We Guarantee Solution Exists?

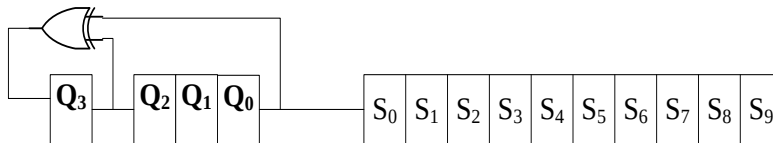
FFT

• Q: What should we do if we find no solution?

♦ 1

♦ 2

♦ 3



$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Q_3 \\ Q_2 \\ Q_1 \\ Q_0 \end{bmatrix} = \begin{bmatrix} S_0 = 1 \\ S_1 = X \\ S_2 = X \\ S_3 = X \\ S_4 = 0 \\ S_5 = 1 \\ S_6 = X \\ S_7 = X \\ S_8 = 1 \\ S_9 = 0 \end{bmatrix}$$